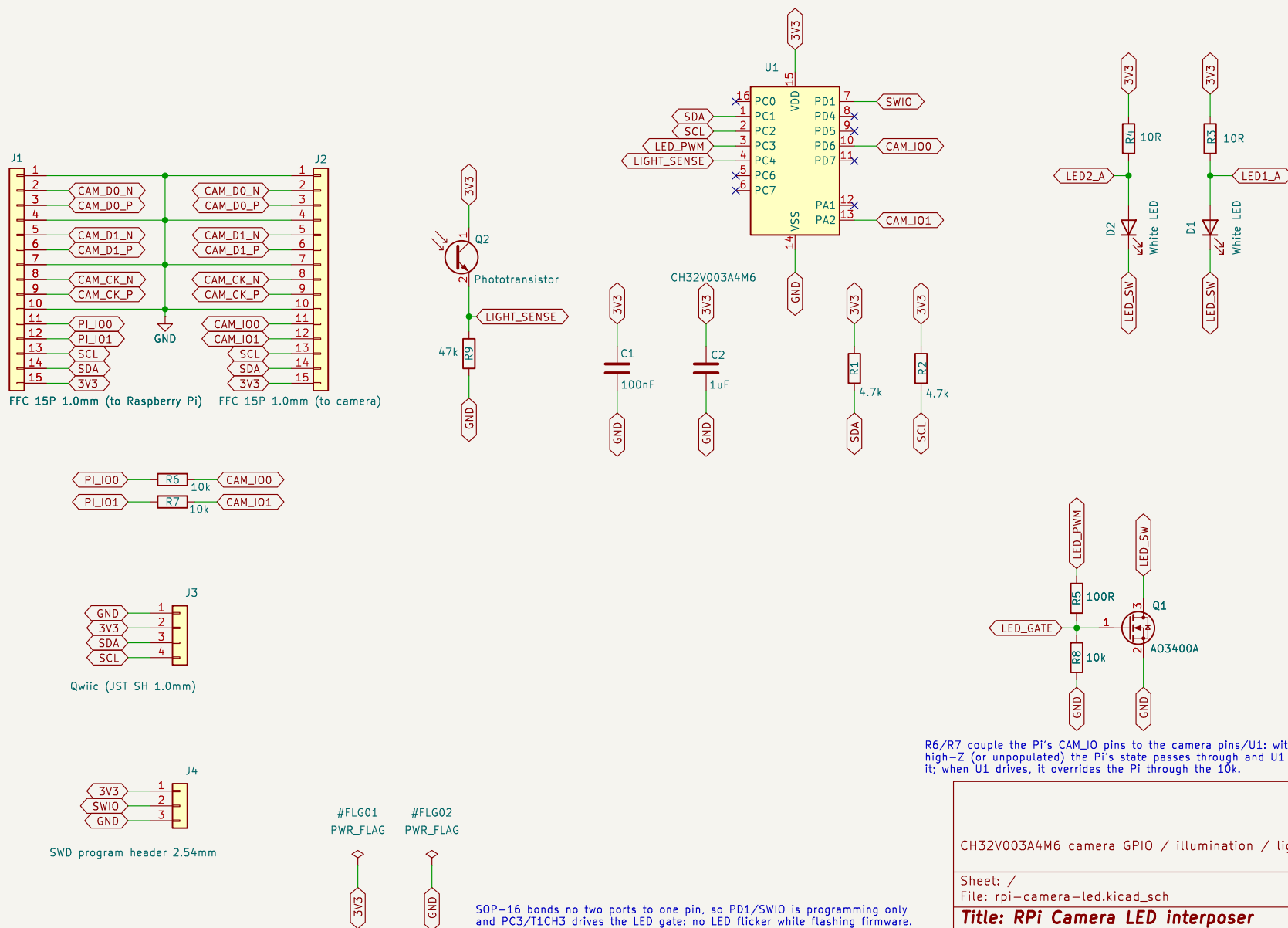


RPi Camera LED interposer: sits between a Raspberry Pi and its camera.
MIPI lanes, I2C, GND and 3V3 pass straight through J1 -> J2.
U1 (I2C slave on the camera bus) drives the camera GPIO pins 11/12,
PWM-drives the illumination LEDs and reads ambient light on its ADC.

LED current is set by R3/R4 against the 3V3 rail (about 30-40mA each with $V_f=2.9V$). Total draw comes from the Pi camera 3V3 supply - keep the duty/current within the Pi's camera connector budget.



R6/R7 couple the Pi's CAM_IO pins to the camera pins/U1: with U1 high-Z (or unpopulated) the Pi's state passes through and U1 can read it; when U1 drives, it overrides the Pi through the 10k.

CH32V003A4M6 camera GPIO / illumination / light sensor

Sheet: /
File: rpi-camera-led.kicad_sch

Title: RPi Camera LED interposer

Size: A4 Date: 2026-09-01

KiCad E.D.A. 9.0.7

Rev: A

Id: 1/1

All parts chosen for JLCPCB / NextPCB Rev0 assembly; see research/*.md for availability and pricing (checked 2026-09-01).

SOP-16 bonds no two ports to one pin, so PD1/SWIO is programming only and PC3/T1CH3 drives the LED gate: no LED flicker while flashing firmware. PC6, PC7, PD4, PD5, PD7/NRST, PA1 and PC0 are spare.